

Supplementary Material

Table S1: Inclusion Criteria for Paper Selection Based on Thematic Scoring

Ref.	Focus on LLM Agentic Behavior	Training Mechanisms for Autonomy	Interaction with External Tools or Environments	Frameworks for Single/Multi-Agent Systems	Reasoning, Planning, Memory, or Self-Reflection	Prompting, Fine-Tuning, or Memory Augmentation	Evaluation Metrics and Benchmarks for Agentic Performance	Challenges, Limitations, and Ethical Considerations	Total Score
[45]	8	8	8	8	8	8	8	6	62
[4]	6	8	8	8	7	7	6	6	56
[28]	8	8	8	8	8	8	7	8	63
[49]	8	8	8	8	8	8	8	5	61
[11]	8	8	8	7	7	5	6	8	57
[56]	5	5	8	6	8	8	7	8	55
[75]	8	8	8	8	8	8	7	8	63
[40]	6	7	7	8	7	6	8	6	55
[48]	8	7	8	5	5	6	7	8	54
[37]	8	8	8	8	8	8	8	6	62
[99]	6	7	8	8	8	5	5	7	54
[103]	8	8	6	5	6	8	7	5	53
[102]	8	8	8	8	8	8	8	7	63
[54]	6	5	7	8	8	6	8	7	55
[89]	8	8	8	8	8	8	8	5	61
[97]	7	7	6	8	7	8	5	6	54
[69]	6	7	7	5	7	8	8	8	56
[96]	8	8	8	8	8	8	7	8	63
[38]	6	6	6	7	5	8	8	8	54
[63]	8	8	7	6	6	8	5	7	55
[46]	8	8	8	8	8	8	8	6	62
[16]	6	5	7	7	6	8	7	7	53
[19]	8	8	8	8	8	8	8	7	63
[90]	5	8	6	7	6	8	8	8	56
[52]	8	6	7	5	6	7	7	8	54
[108]	8	8	8	8	8	8	8	5	61

Table S1 – continued from previous page

Ref.	Focus on LLM Agentic Behavior	Training Mechanisms for Autonomy	Interaction with External Tools or Environments	Frameworks for Single/Multi-Agent Systems	Reasoning, Planning, Memory, or Self-Reflection	Prompting, Fine-Tuning, or Memory Augmentation	Evaluation Metrics and Benchmarks for Agentic Performance	Challenges, Limitations, and Ethical Considerations	Total Score
[13]	5	5	6	8	7	8	8	8	55
[93]	7	6	8	8	7	7	6	5	54
[60]	6	7	7	8	8	6	6	8	56
[33]	8	8	8	8	8	8	8	6	62
[70]	7	8	8	8	6	5	8	6	56
[67]	6	6	8	5	8	8	7	8	56
[81]	8	8	8	8	8	8	8	7	63
[30]	5	8	5	6	7	8	8	7	54
[73]	6	6	8	6	7	5	8	8	54
[14]	8	8	8	8	8	8	8	5	61
[68]	6	8	7	8	8	6	7	7	57
[26]	7	6	5	6	7	8	8	8	55
[88]	8	8	8	8	8	8	8	6	62
[83]	6	8	7	8	7	8	8	5	57
[39]	7	8	6	7	8	5	8	7	56
[59]	5	6	8	8	6	7	8	7	55
[91]	8	8	8	8	8	8	8	7	63
[25]	6	8	8	6	8	6	7	8	57
[2]	5	7	6	8	8	7	6	7	54
[41]	7	5	7	8	8	6	7	6	54
[58]	8	8	8	8	8	8	8	6	62
[44]	8	7	7	8	8	7	5	6	56
[95]	7	8	6	8	8	7	7	5	56
[66]	6	7	5	8	6	8	7	8	55
[43]	8	6	8	7	8	5	7	6	55
[53]	8	8	8	8	8	8	8	6	62
[34]	5	6	8	8	7	6	8	7	55
[9]	8	8	8	8	8	8	8	5	61
[47]	7	8	6	8	8	5	7	8	57
[3]	8	8	8	8	8	8	8	6	62
[106]	5	6	7	8	8	7	6	8	55
[76]	8	7	8	8	6	8	8	5	58

Table S1 – continued from previous page

Ref.	Focus on LLM Agentic Behavior	Training Mechanisms for Autonomy	Interaction with External Tools or Environments	Frameworks for Single/Multi-Agent Systems	Reasoning, Planning, Memory, or Self-Reflection	Prompting, Fine-Tuning, or Memory Augmentation	Evaluation Metrics and Benchmarks for Agentic Performance	Challenges, Limitations, and Ethical Considerations	Total Score
[105]	6	8	8	8	8	7	8	6	59
[27]	8	8	8	8	8	8	7	8	63
[12]	5	5	6	7	8	8	8	7	54
[5]	7	8	8	8	8	6	8	7	60
[78]	8	8	8	8	8	8	8	5	61
[98]	6	7	8	5	8	7	8	8	57
[79]	8	8	7	8	8	8	8	6	61
[23]	5	8	8	6	7	8	8	8	58
[17]	8	8	8	8	8	8	8	7	63
[85]	7	6	5	8	8	8	7	8	57
[64]	8	8	8	8	8	8	6	8	62
[65]	6	8	7	8	5	8	7	8	57
[100]	8	8	8	8	8	8	8	5	61
[21]	5	7	8	8	6	8	8	7	57
[8]	8	8	8	8	8	8	7	8	63
[36]	8	6	8	7	8	8	8	5	58
[71]	8	8	8	8	8	8	8	6	62
[57]	7	8	5	8	8	6	8	8	58
[94]	8	8	8	8	8	8	8	7	63
[7]	6	5	8	8	7	8	8	8	58
[35]	8	8	8	8	8	8	6	8	62
[107]	8	7	8	5	8	8	8	8	60
[61]	8	8	8	8	8	8	8	5	61
[50]	5	8	6	8	8	7	8	8	58
[6]	8	8	8	8	8	8	7	8	63
[86]	7	6	8	8	8	5	8	8	58
[55]	8	8	8	8	8	8	8	6	62
[101]	8	5	7	8	8	8	6	8	58
[10]	8	8	8	8	8	8	8	7	63
[22]	6	8	8	7	8	8	5	8	58
[18]	8	8	8	8	8	8	6	8	62
[32]	5	8	8	8	6	8	8	7	58

Table S1 – continued from previous page

Ref.	Focus on LLM Agentic Behavior	Training Mechanisms for Autonomy	Interaction with External Tools or Environments	Frameworks for Single/Multi- Agent Systems	Reasoning, Planning, Memory, or Self-Reflection	Prompting, Fine-Tuning, or Memory Augmentation	Evaluation Metrics and Benchmarks for Agentic Performance	Challenges, Limitations, and Ethical Considerations	Total Score
[77]	8	8	8	8	8	8	8	5	61
[80]	7	8	8	5	8	8	8	8	60
[104]	8	8	8	8	8	8	7	8	63
[92]	8	6	8	8	8	5	8	8	59
[51]	8	8	8	8	8	8	8	6	62
[72]	5	7	8	8	8	8	6	8	58
[31]	5	8	5	6	7	8	8	7	54
[42]	6	6	8	6	7	5	8	8	54
[15]	8	8	8	8	8	8	8	5	61
[29]	6	8	7	8	8	6	7	7	57
[74]	7	6	5	6	7	8	8	8	55
[82]	8	8	8	8	8	8	8	6	62
[84]	8	8	8	8	8	8	8	5	61
[87]	5	7	8	8	6	8	8	7	57
[20]	8	8	8	8	8	8	7	8	63
[24]	8	6	8	7	8	8	8	5	58
[1]	7	8	5	8	8	6	8	8	58
[62]	5	6	7	8	5	6	7	8	52

Table S2: Reasoning, planning, and memory techniques in LLM-based agents across single-agent, multi-agent, and widely-used categories

Reasoning techniques in LLM-based agents	
Category	Technique/Framework
Single-agent reasoning	TOOLLLM depth-first search-based decision tree [56], EASYTOOL task decomposition [90], MATRIX simulation-based self-critique [52], ToolEmu risk-prompted reasoning [59], ToT [2], CogMir social-context prompts [44], Moral regret mechanisms [66], SHAP-based interpretability [100], Beneficial hallucination-driven hypothesis testing [15], ChatNT perplexity-informed confidence estimation [47]
Multi-agent reasoning	Prompt-Structured Strategic Reasoning [41], AGENTVERSE multi-agent collaborative reasoning [11], Theory-of-Mind (ToM) inspired belief module [37]
Widely used techniques	CoT [2, 4, 23, 37, 42, 44, 59, 72, 76, 82, 84, 85, 89, 92, 93, 98, 108], ReAct [16, 61, 81, 99, 102], Self-Reflection and Self-Critique [5–7, 44, 47, 50, 57, 61, 71, 85, 92, 95, 96, 98, 102], Self-Consistency methods [33, 70], Multi-Agent Collaborative Reasoning [12, 20, 46, 96], Ensemble and consensus strategies [18, 33, 48]
Planning techniques in LLM-based agents	
Category	Technique/Framework
Single-agent planning	TIMEARENA heuristic planning [102], ReHAC multi-step planning [16], Task stack-based planning [108], Single-shot planning [9], World Knowledge Model guided planning [55], RAFA multi-step trajectory generation [45]
Multi-agent planning	ProAgent modularized planning [93], Plan-first execution paradigm [26], DeCoAgent few-shot prompt-based task decomposition [27], Causal strategy learning systems [35], Star-Agents evolutionary strategy [107], UrbanKGent fixed pipeline [50], SMART-LLM coalition formation [31], Human-in-the-loop workflow [5]
Widely used techniques	Multi-step planning [9, 16, 31, 45], Task decomposition planning [27, 74, 90, 102], ReAct-based planning [62, 99, 102], Reflexion planning [16, 62, 102], Re-planning and dynamic adaptation [64, 71], Plan generation and evaluation [25, 69]
Memory techniques in LLM-based agents	
Category	Technique/Framework
Single-agent memory	RE-Bench [75], RCAgent’s OBSK [70], Memory buffers [14], VLN-Imagine [53], ToolEmu prompt context tracking [59], MAgentBench compressed Research Logs [25], Formal Reasoning Graphs [2], STEVE-EYE episodic re-prompting [106], Moral alignment prompts [66], AGENT-BENCH extended histories [43], CogAgent external screenshots and latent weights [23], Rolling logs for robots [17], Prior tool outputs in context [85], Negotiated ontologies [100], Concatenated dialogue contexts [71]
Multi-agent memory	ProAgent recent-K and relevant-K retrieval [93], ColaCare retrieval-augmented generation [73], Satellite network knowledge bases [98], Re-feeding prior dialogue turns [44], Discounted back-tracking [34], MATRIX running textual memory [52], DeCoAgent persistent long-term storage [27], CompeteAI daybook and comment histories [105]
Widely used techniques	Context window memory [8, 21, 29, 31, 50, 51, 57, 62, 64, 72, 80, 94], Conversational and dialogue history [5, 12, 20, 29, 36, 69, 74], Reflexion-style memory systems [7, 16, 61, 63], RAG methods [5, 21, 27, 73], Working memory and scratchpad [2, 9, 63], Long-term episodic memory [87, 94], Short-term memory optimization [7, 24, 29, 82]

Table S3: Identified research gaps and potential research paths for advancing LLM-based agent systems.

Research Gaps	Potential Research Paths
Lack of Verifiable Reasoning	1. Replace unstructured chain-of-thought with logically consistent and auditable reasoning frameworks. 2. Integrate symbolic AI approaches (e.g., BDI agents) for verifiable decision-making. 3. Ground reasoning in domain-specific processes for technical and scientific accuracy. 4. Incorporate fine-grained error detection and correction mechanisms during inference.
Limited Self-Improvement	1. Employ introspection-based reinforcement learning for continuous error identification and correction. 2. Use cooperative multi-agent evolution to share and refine learned strategies. 3. Maintain adaptive capabilities without requiring full-scale retraining. 4. Leverage reflective feedback loops to iteratively enhance performance.
Infrastructure Bottlenecks	1. Optimize inference latency through KV-caching and low-latency decoding pipelines. 2. Implement lightweight, modular architectures for resource-constrained settings. 3. Enable efficient on-device and streaming computation for real-time tasks.
Weak Multi-Agent Communication	1. Design adaptive dialogue protocols inspired by human social cognition. 2. Integrate contextual modeling for dynamic, multi-agent negotiation and coordination. 3. Improve real-time collaboration in competitive and cooperative environments.
Limited Context-Sensitive Collaboration	1. Develop proactive goal refinement methods for dynamic user needs. 2. Enhance co-creative interaction to anticipate and adapt to evolving task requirements. 3. Move beyond passive instruction-following toward proactive task engagement.
Insufficient Personalization	1. Build persistent, evolving user models based on long-term interactions. 2. Incorporate privacy-preserving memory architectures. 3. Adapt multi-modal interfaces dynamically to user preferences.
Vulnerability to Adversarial Triggers	1. Establish robust detection and prevention mechanisms against backdoor attacks triggered by contextual or environmental signals. 2. Implement adversarial robustness testing during training and deployment. 3. Continuously monitor and respond to unsafe emergent behaviors.
Inadequate Explainability	1. Embed explainability directly into the reasoning process. 2. Provide step-by-step justifications for decisions in human-readable form. 3. Tailor explanations to different levels of user expertise.
Incomplete Evaluation Frameworks	1. Establish standardized benchmarks for multi-agent reasoning in complex contexts. 2. Include negotiation-heavy, ambiguous, and competitive tasks in evaluation. 3. Measure adaptability, reliability, and fairness across diverse scenarios.
Neglected Tool Adaptation	1. Treat agent tools as learnable and optimizable parameters. 2. Automate prompt engineering for task-specific and tool-specific optimization. 3. Evaluate the effect of tool adaptation on performance across varied tasks.

References

- [1] Mahyar Abbasian, Iman Azimi, Amir M Rahmani, and Ramesh Jain. Conversational health agents: a personalized large language model-powered agent framework. *JAMIA Open*, 8(4):ooaf067, 2025.
- [2] Samir Abdaljalil, Hasan Kurban, Khalid Qaraqe, and Erchin Serpedin. Theorem-of-thought: A multi-agent framework for abductive, deductive, and inductive reasoning in language models. In Yuji Zhang, Canyu Chen, Sha Li, Mor Geva, Chi Han, Xiaozhi Wang, Shangbin Feng, Silin Gao, Isabelle Augenstein, Mohit Bansal, Manling Li, and Heng Ji, editors, *Proceedings of the 3rd Workshop on Towards Knowledgeable Foundation Models (KnowFM)*, pages 111–119, Vienna, Austria, August 2025. Association for Computational Linguistics. ISBN 979-8-89176-283-1. URL <https://aclanthology.org/2025.knowllm-1.10/>.
- [3] Dilip Arumugam and Thomas L. Griffiths. Toward efficient exploration by large language model agents. In *The Exploration in AI Today Workshop at ICML 2025*, 2025. URL <https://openreview.net/forum?id=7cx0WZs7LY>.
- [4] Xingyuan Bai, Shaobin Huang, Chi Wei, and Rui Wang. Collaboration between intelligent agents and large language models: A novel approach for enhancing code generation capability. *Expert Systems with Applications*, 269:126357, 2025.
- [5] Federico Lorenzo Barra, Giovanna Rodella, Alessandro Costa, Antonio Scalogna, Luca Carenzo, Alice Monzani, and Francesco Della Corte. From prompt to platform: an agentic ai workflow for healthcare simulation scenario design. *Advances in Simulation*, 10(1):29, 2025.
- [6] Nikhil Behari, Edwin Zhang, Yunfan Zhao, Aparna Taneja, Dheeraj Nagaraj, and Milind Tambe. A decision-language model (dlm) for dynamic restless multi-armed bandit tasks in public health. *Advances in Neural Information Processing Systems*, 37:3964–4002, 2024.
- [7] Xiaohe Bo, Zeyu Zhang, Quanyu Dai, Xueyang Feng, Lei Wang, Rui Li, Xu Chen, and Ji-Rong Wen. Reflective multi-agent collaboration based on large language models. *Advances in Neural Information Processing Systems*, 37:138595–138631, 2024.
- [8] Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023.
- [9] Yifu Cai, Xinyu Li, Mononito Goswami, Michał Wiliński, Gus Welter, and Artur Dubrawski. LLM agents struggle at time series machine learning engineering. In *1st ICML Workshop on Foundation Models for Structured Data*, 2025. URL <https://openreview.net/forum?id=04fwptRMB>.
- [10] Ma Chang, Junlei Zhang, Zhihao Zhu, Cheng Yang, Yujiu Yang, Yaohui Jin, Zhenzhong Lan, Lingpeng Kong, and Junxian He. Agentboard: An analytical evaluation board of multi-turn llm agents. *Advances in neural information processing systems*, 37:74325–74362, 2024.
- [11] Weize Chen, Yusheng Su, Jingwei Zuo, Cheng Yang, Chenfei Yuan, et al. Agentverse: Facilitating multi-agent collaboration and exploring emergent behaviors in agents. 2024.
- [12] Xi Chen, Huahui Yi, Mingke You, WeiZhi Liu, Li Wang, Hairui Li, Xue Zhang, Yingman Guo, Lei Fan, Gang Chen, et al. Enhancing diagnostic capability with multi-agents conversational large language models. *NPJ digital medicine*, 8(1):159, 2025.
- [13] Junyan Cheng and Peter Chin. Sociodojo: Building lifelong analytical agents with real-world text and time series. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=s9z0HzWJJp>.
- [14] Andrea Cuadra, Justine Breuch, Samantha Estrada, David Ihim, Isabelle Hung, Derek Askaryar, Marwan Hassanien, Kristen L Fessele, and James A Landay. Digital forms for all: A holistic multimodal large language model agent for health data entry. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(2):1–39, 2024.
- [15] Robert Feldt, Sungmin Kang, Juyeon Yoon, and Shin Yoo. Towards autonomous testing agents via conversational large language models. In *2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, pages 1688–1693. IEEE, 2023.
- [16] Xueyang Feng, Zhi-Yuan Chen, Yujia Qin, Yankai Lin, Xu Chen, Zhiyuan Liu, and Ji-Rong Wen. Large language model-based human-agent collaboration for complex task solving. In Yaser Al-Onaizan, Mohit Bansal,

- and Yun-Nung Chen, editors, *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1336–1357, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.72. URL <https://aclanthology.org/2024.findings-emnlp.72/>.
- [17] Laurent Frering, Gerald Steinbauer-Wagner, and Andreas Holzinger. Integrating belief-desire-intention agents with large language models for reliable human–robot interaction and explainable artificial intelligence. *Engineering Applications of Artificial Intelligence*, 141:109771, 2025.
- [18] Yao Fu, Dong-Ki Kim, Jaekyeom Kim, Sungryull Sohn, Lajanugen Logeswaran, Kyunghoon Bae, and Honglak Lee. Autoguide: Automated generation and selection of context-aware guidelines for large language model agents. *Advances in Neural Information Processing Systems*, 37:119919–119948, 2024.
- [19] Sonny George, Chris Sypherd, and Dylan Cashman. Probing the capacity of language model agents to operationalize disparate experiential context despite distraction. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15447–15459, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.905. URL <https://aclanthology.org/2024.findings-emnlp.905/>.
- [20] Alireza Ghafarollahi and Markus J Buehler. Protagents: protein discovery via large language model multi-agent collaborations combining physics and machine learning. *Digital Discovery*, 3(7):1389–1409, 2024.
- [21] Alex J Goodell, Simon N Chu, Dara Rouholiman, and Larry F Chu. Large language model agents can use tools to perform clinical calculations. *npj Digital Medicine*, 8(1):163, 2025.
- [22] Niklas Hemken, Sai Koneru, Florian Jacob, Hannes Hartenstein, and Jan Niehues. Can a large language model keep my secrets? a study on LLM-controlled agents. In Jin Zhao, Mingyang Wang, and Zhu Liu, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 746–759, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-254-1. URL <https://aclanthology.org/2025.acl-srw.49/>.
- [23] Wenyi Hong, Weihang Wang, Qingsong Lv, Jiazheng Xu, Wenmeng Yu, Junhui Ji, Yan Wang, Zihan Wang, Yuxiao Dong, Ming Ding, et al. Cogagent: A visual language model for gui agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14281–14290, 2024.
- [24] Huazhen Huang, Xianguo Shi, Hongyang Lei, Fan Hu, and Yunpeng Cai. Protchat: An ai multi-agent for automated protein analysis leveraging gpt-4 and protein language model. *Journal of Chemical Information and Modeling*, 65(1):62–70, 2024.
- [25] Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. Benchmarking large language models as ai research agents. In *NeurIPS 2023 Foundation Models for Decision Making Workshop*, 2023.
- [26] Xu Huang, Jianxun Lian, Yuxuan Lei, Jing Yao, Defu Lian, and Xing Xie. Recommender ai agent: Integrating large language models for interactive recommendations. *ACM Trans. Inf. Syst.*, 43(4), June 2025. ISSN 1046-8188. doi: 10.1145/3731446. URL <https://doi.org/10.1145/3731446>.
- [27] Anan Jin, Yuhang Ye, Brian Lee, and Yuansong Qiao. Decoagent: Large language model empowered decentralized autonomous collaboration agents based on smart contracts. *IEEE Access*, 2024.
- [28] Yuping Jin and Jun Ma. Large language model as parking planning agent in the context of mixed period of autonomous vehicles and human-driven vehicles. *Sustainable Cities and Society*, 117:105940, 2024.
- [29] Eunkyung Jo, Daniel A. Epstein, Hyunhoon Jung, and Young-Ho Kim. Understanding the benefits and challenges of deploying conversational ai leveraging large language models for public health intervention. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI ’23, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9781450394215. doi: 10.1145/3544548.3581503. URL <https://doi.org/10.1145/3544548.3581503>.
- [30] Steven Jorgensen, Giorgia Nadizar, Gloria Pietropolli, Luca Manzoni, Eric Medvet, Una-May O’Reilly, and Erik Hemberg. Large language model-based test case generation for gp agents. In *Proceedings of the genetic and evolutionary computation conference*, pages 914–923, 2024.
- [31] Shyam Sundar Kannan, Vishnunandan L. N. Venkatesh, and Byung-Cheol Min. Smart-llm: Smart multi-agent robot task planning using

- large language models. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 12140–12147, 2024. doi: 10.1109/IROS58592.2024.10802322.
- [32] Mohammad Afrah Khan, Mahsa Amani, Soumi Das, Bishwamittra Ghosh, Qinyuan Wu, Krishna P. Gummadi, Manish Gupta, and Abhilasha Ravichander. In agents we trust, but who do agents trust? latent source preferences steer LLM generations. In *ICML 2025 Workshop on Reliable and Responsible Foundation Models*, 2025. URL <https://openreview.net/forum?id=oZMisPPggL>.
- [33] Yubin Kim, Chanwoo Park, Hyewon Jeong, Yik S Chan, Xuhai Xu, Daniel McDuff, Hyeonhoon Lee, Marzyeh Ghassemi, Cynthia Breazeal, and Hae W Park. Mdagents: An adaptive collaboration of llms for medical decision-making. *Advances in Neural Information Processing Systems*, 37: 79410–79452, 2024.
- [34] Lars Henning Klein, Nearchos Potamitis, Roland Aydin, Robert West, Caglar Gulcehre, and Akhil Arora. Fleet of agents: Coordinated problem solving with large language models. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=yNpYb376zf>.
- [35] Andrew Lampinen, Stephanie Chan, Ishita Dasgupta, Andrew Nam, and Jane Wang. Passive learning of active causal strategies in agents and language models. *Advances in Neural Information Processing Systems*, 36: 1283–1297, 2023.
- [36] Guohao Li, Hasan Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. Camel: Communicative agents for "mind" exploration of large language model society. *Advances in Neural Information Processing Systems*, 36: 51991–52008, 2023.
- [37] Huao Li, Yu Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Charles Lewis, and Katia Sycara. Theory of mind for multi-agent collaboration via large language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 180–192, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.13. URL <https://aclanthology.org/2023.emnlp-main.13/>.
- [38] Nian Li, Chen Gao, Mingyu Li, Yong Li, and Qingmin Liao. EconAgent: Large language model-empowered agents for simulating macroeconomic activities. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15523–15536, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.829. URL <https://aclanthology.org/2024.acl-long.829/>.
- [39] Ruosen Li, Ziming Luo, and Xinya Du. Fine-grained hallucination detection and mitigation in language model mathematical reasoning, 2024. URL <https://openreview.net/forum?id=7bwE5MJAVJ>.
- [40] Xiang Lisa Li, Neil Chowdhury, Daniel D. Johnson, Tatsunori Hashimoto, Percy Liang, Sarah Schwettmann, and Jacob Steinhardt. Eliciting language model behaviors with investigator agents. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=AulTigiaMv>.
- [41] Zhemin Li, Ruobing Zhang, Zhengming Wang, Zheng Xie, and Yiping Song. Llm-guided decision-making toolkit for multi-agent reinforcement learning. *Neurocomputing*, 638:130105, 2025.
- [42] Wei Liu, Jun Li, Yitao Tang, Yining Zhao, Chaozhong Liu, Meiyi Song, Zhenlin Ju, Shwetha V Kumar, Yiling Lu, Rehan Akbani, et al. Drbioright 2.0: an llm-powered bioinformatics chatbot for large-scale cancer functional proteomics analysis. *Nature communications*, 16(1):2256, 2025.
- [43] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. Agentbench: Evaluating LLMs as agents. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=zAdUB0aCTQ>.
- [44] Xuan Liu, Jie ZHANG, HaoYang Shang, Song Guo, Yang Chengxu, and Quanyan Zhu. Exploring prosocial irrationality for LLM agents: A social cognition view. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=u8VOQVzduP>.

- [45] Zhihan Liu, Hao Hu, Shenao Zhang, Hongyi Guo, Shuqi Ke, Boyi Liu, and Zhaoran Wang. Reason for future, act for now: a principled architecture for autonomous llm agents. In *Proceedings of the 41st International Conference on Machine Learning*, ICML’24. JMLR.org, 2024.
- [46] Meng Lu, Brandon Ho, Dennis Ren, and Xuan Wang. Triageagent: Towards better multi-agents collaborations for large language model-based clinical triage. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 5747–5764, 2024.
- [47] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024.
- [48] Hao Ma, Tianyi Hu, Zhiqiang Pu, Boyin Liu, Xiaolin Ai, Yanyan Liang, and Min Chen. Coevolving with the other you: Fine-tuning LLM with sequential cooperative multi-agent reinforcement learning. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=0o0CoZFVK3>.
- [49] Bo Ni and Markus J Buehler. Mechagents: Large language model multi-agent collaborations can solve mechanics problems, generate new data, and integrate knowledge. *Extreme Mechanics Letters*, 67:102131, 2024.
- [50] Yansong Ning and Hao Liu. Urbankgent: A unified large language model agent framework for urban knowledge graph construction. *Advances in Neural Information Processing Systems*, 37:123127–123154, 2024.
- [51] Jing-Cheng Pang, Si-Hang Yang, Kaiyuan Li, Jiaji Zhang, Xiong-Hui Chen, Nan Tang, and Yang Yu. Kalm: Knowledgeable agents by offline reinforcement learning from large language model rollouts. *Advances in Neural Information Processing Systems*, 37:126620–126652, 2024.
- [52] Xianghe Pang, Shuo Tang, Rui Ye, Yuxin Xiong, Bolun Zhang, Yanfeng Wang, and Siheng Chen. Self-alignment of large language models via monopolylogue-based social scene simulation. In *Proceedings of the 41st International Conference on Machine Learning*, ICML’24. JMLR.org, 2024.
- [53] Akhil Perincherry, Jacob Krantz, and Stefan Lee. Do visual imaginations improve vision-and-language navigation agents? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3846–3855, June 2025.
- [54] Cheng Qian, Bingxiang He, Zhong Zhuang, Jia Deng, Yujia Qin, Xin Cong, Zhong Zhang, Jie Zhou, Yankai Lin, Zhiyuan Liu, and Maosong Sun. Tell me more! towards implicit user intention understanding of language model driven agents. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1088–1113, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.61. URL <https://aclanthology.org/2024.acl-long.61/>.
- [55] Shuofei Qiao, Runnan Fang, Ningyu Zhang, Yuqi Zhu, Xiang Chen, Shumin Deng, Yong Jiang, Pengjun Xie, Fei Huang, and Huajun Chen. Agent planning with world knowledge model. *Advances in Neural Information Processing Systems*, 37:114843–114871, 2024.
- [56] Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, dahai li, Zhiyuan Liu, and Maosong Sun. ToolLLM: Facilitating large language models to master 16000+ real-world APIs. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=dHng200Jjr>.
- [57] Yuxiao Qu, Tianjun Zhang, Naman Garg, and Aviral Kumar. Recursive introspection: Teaching language model agents how to self-improve. *Advances in Neural Information Processing Systems*, 37:55249–55285, 2024.
- [58] Samuel Rothfarb, Mikayla Friday, Xingyu Wang, Arash Zaghi, and Baikun Li. Multi-agent large language model frameworks: Unlocking new possibilities for optimizing wastewater treatment operation. *Environmental Research*, 275:121401, 2025.
- [59] Yangjun Ruan, Honghua Dong, Andrew Wang, Silviu Pitis, Yongchao Zhou, Jimmy Ba, Yann Dubois, Chris J. Maddison, and Tatsunori Hashimoto. Identifying the risks of LM agents with an LM-emulated sandbox. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=GEcwtMk1uA>.
- [60] Raphael Schumann, Wanrong Zhu, Weixi Feng, Tsu-Jui Fu, Stefan Riezler, and William Yang Wang. Velma: Verbalization embodiment of llm agents for vision and language navigation in street view. In *Proceedings of the*

- AAAI Conference on Artificial Intelligence*, volume 38, pages 18924–18933, 2024.
- [61] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 36:8634–8652, 2023.
 - [62] Harmanpreet Singh, Nikhil Verma, Yixiao Wang, Manasa Bharadwaj, Homa Fashandi, Kevin Ferreira, and Chul Lee. Personal large language model agents: A case study on tailored travel planning. In Franck Dernoncourt, Daniel Preotiu-Pietro, and Anastasia Shimorina, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 486–514, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-industry.37. URL <https://aclanthology.org/2024.emnlp-industry.37/>.
 - [63] Harmanpreet Singh, Nikhil Verma, Yixiao Wang, Manasa Bharadwaj, Homa Fashandi, Kevin Ferreira, and Chul Lee. Personal large language model agents: A case study on tailored travel planning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 486–514, 2024.
 - [64] Chan Hee Song, Jiaman Wu, Clayton Washington, Brian M Sadler, Wei-Lun Chao, and Yu Su. Llm-planner: Few-shot grounded planning for embodied agents with large language models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2998–3009, 2023.
 - [65] Yu Su, Diyi Yang, Shunyu Yao, and Tao Yu. Language agents: Foundations, prospects, and risks. In Jessy Li and Fei Liu, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Tutorial Abstracts*, pages 17–24, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-tutorials.3. URL <https://aclanthology.org/2024.emnlp-tutorials.3/>.
 - [66] Elizaveta Tennant, Stephen Hailes, and Mirco Musolesi. Moral alignment for LLM agents. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=MeGDmZjUXy>.
 - [67] Jen tse Huang, Jiaxu Zhou, Tailin Jin, Xuhui Zhou, Zixi Chen, Wenxuan Wang, Youliang Yuan, Michael Lyu, and Maarten Sap. On the resilience of LLM-based multi-agent collaboration with faulty agents. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=bkiM54QftZ>.
 - [68] Lei Wang, Jingsen Zhang, Hao Yang, Zhi-Yuan Chen, Jiakai Tang, Zeyu Zhang, Xu Chen, Yankai Lin, Hao Sun, Ruihua Song, Xin Zhao, Jun Xu, Zhicheng Dou, Jun Wang, and Ji-Rong Wen. User behavior simulation with large language model-based agents. *ACM Trans. Inf. Syst.*, 43(2), January 2025. ISSN 1046-8188. doi: 10.1145/3708985. URL <https://doi.org/10.1145/3708985>.
 - [69] Ruiyi Wang, Haofei Yu, Wenxin Zhang, Zhengyang Qi, Maarten Sap, Yonatan Bisk, Graham Neubig, and Hao Zhu. SOTOPIA- π : Interactive learning of socially intelligent language agents. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12912–12940, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.698. URL <https://aclanthology.org/2024.acl-long.698/>.
 - [70] Zefan Wang, Zichuan Liu, Yingying Zhang, Aoxiao Zhong, Jihong Wang, Fengbin Yin, Lunting Fan, Lingfei Wu, and Qingsong Wen. Ragent: Cloud root cause analysis by autonomous agents with tool-augmented large language models. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24*, page 4966–4974, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400704369. doi: 10.1145/3627673.3680016. URL <https://doi.org/10.1145/3627673.3680016>.
 - [71] Zihao Wang, Shaofei Cai, Guanzhou Chen, Anji Liu, Xiaojian Ma, Yitao Liang, and Team CraftJarvis. Describe, explain, plan and select: interactive planning with large language models enables open-world multi-task agents. In *Proceedings of the 37th International Conference on Neural Information Processing Systems, NIPS '23*, Red Hook, NY, USA, 2023. Curran Associates Inc.
 - [72] Zihao Wang, Shaofei Cai, Zhancun Mu, Haowei Lin, Ceyao Zhang, Xuejie Liu, Qing Li, Anji Liu, Xiaojian Shawn Ma, and Yitao Liang. Omnijarvis: Unified vision-language-action tokenization enables open-world instruction

following agents. *Advances in Neural Information Processing Systems*, 37: 73278–73308, 2024.

- [73] Zixiang Wang, Yinghao Zhu, Huiya Zhao, Xiaochen Zheng, Dehao Sui, Tianlong Wang, Wen Tang, Yasha Wang, Ewen Harrison, Chengwei Pan, Junyi Gao, and Liantao Ma. Colacare: Enhancing electronic health record modeling through large language model-driven multi-agent collaboration. In *Proceedings of the ACM on Web Conference 2025*, WWW '25, page 2250–2261, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400712746. doi: 10.1145/3696410.3714877. URL <https://doi.org/10.1145/3696410.3714877>.
- [74] Yuxi Wei, Zi Wang, Yifan Lu, Chenxin Xu, Changxing Liu, Hao Zhao, Siheng Chen, and Yanfeng Wang. Editable scene simulation for autonomous driving via collaborative llm-agents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15077–15087, 2024.
- [75] Hjalmar Wijk, Tao Roa Lin, Joel Becker, Sami Jawhar, Neev Parikh, Thomas Broadley, Lawrence Chan, Michael Chen, Joshua M Clymer, Jai Dhyani, Elena Elicheva, Katharyn Garcia, Brian Goodrich, Nikola Jurkovic, Megan Kinniment, Aron Lajko, Seraphina Nix, Lucas Jun Koba Sato, William Saunders, Maksym Taran, Ben West, and Elizabeth Barnes. RE-bench: Evaluating frontier AI r&d capabilities of language model agents against human experts. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=3rB0bVU6z6>.
- [76] Haoyuan Wu, Zhuolun He, Xinyun Zhang, Xufeng Yao, Su Zheng, Haisheng Zheng, and Bei Yu. Chateda: A large language model powered autonomous agent for eda. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 43(10):3184–3197, 2024.
- [77] Shirley Wu, Shiyu Zhao, Qian Huang, Kexin Huang, Michihiro Yasunaga, Kaidi Cao, Vassilis Ioannidis, Karthik Subbian, Jure Leskovec, and James Y Zou. Avatar: Optimizing llm agents for tool usage via contrastive reasoning. *Advances in Neural Information Processing Systems*, 37:25981–26010, 2024.
- [78] Yuchen Xia, Zhewen Xiao, Nasser Jazdi, and Michael Weyrich. Generation of asset administration shell with large language model agents: Toward semantic interoperability in digital twins in the context of industry 4.0. *IEEE Access*, 12:84863–84877, 2024.
- [79] Zhen Xiang, Linzhi Zheng, Yanjie Li, Junyuan Hong, Qinbin Li, Han Xie, Jiawei Zhang, Zidi Xiong, Chulin Xie, Nathaniel D. Bastian, Carl Yang, Dawn Song, and Bo Li. Guardagent: Safeguard LLM agents via knowledge-enabled reasoning. In *ICML 2025 Workshop on Computer Use Agents*, 2025. URL <https://openreview.net/forum?id=ITuuEaXcSB>.
- [80] Chengxing Xie, Canyu Chen, Feiran Jia, Ziyu Ye, Shiyang Lai, Kai Shu, Jindong Gu, Adel Bibi, Ziniu Hu, David Jurgens, et al. Can large language model agents simulate human trust behavior? *Advances in neural information processing systems*, 37:15674–15729, 2024.
- [81] Mingzhe Xing, Rongkai Zhang, Hui Xue, Qi Chen, Fan Yang, and Zhen Xiao. Understanding the weakness of large language model agents within a complex android environment. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '24, page 6061–6072, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400704901. doi: 10.1145/3637528.3671650. URL <https://doi.org/10.1145/3637528.3671650>.
- [82] Minrui Xu, Dusit Niyato, Jiawen Kang, Zehui Xiong, Shiwen Mao, Zhu Han, Dong In Kim, and Khaled B Letaief. When large language model agents meet 6g networks: Perception, grounding, and alignment. *IEEE Wireless Communications*, 2024.
- [83] Yiheng Xu, Hongjin SU, Chen Xing, Boyu Mi, Qian Liu, Weijia Shi, Binyuan Hui, Fan Zhou, Yitao Liu, Tianbao Xie, Zhoujun Cheng, Siheng Zhao, Lingpeng Kong, Bailin Wang, Caiming Xiong, and Tao Yu. Lemur: Harmonizing natural language and code for language agents. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=hNhwSmtXRh>.
- [84] Xiangyuan Xue, Zeyu Lu, Di Huang, Zidong Wang, Wanli Ouyang, and Lei Bai. Comfybench: Benchmarking llm-based agents in comfyui for autonomously designing collaborative ai systems. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 24614–24624, 2025.
- [85] Jianing Yang, Xuweiyi Chen, Shengyi Qian, Nikhil Madaan, Madhavan Iyengar, David F Fouhey, and Joyce Chai. Llm-grounder: Open-vocabulary 3d visual grounding with large language model as an agent. In

2024 *IEEE International Conference on Robotics and Automation (ICRA)*, pages 7694–7701. IEEE, 2024.

- [86] Wenkai Yang, Xiaohan Bi, Yankai Lin, Sishuo Chen, Jie Zhou, and Xu Sun. Watch out for your agents! investigating backdoor threats to llm-based agents. *Advances in Neural Information Processing Systems*, 37:100938–100964, 2024.
- [87] Yijun Yang, Tianyi Zhou, Kanxue Li, Dapeng Tao, Lusong Li, Li Shen, Xiaodong He, Jing Jiang, and Yuhui Shi. Embodied multi-modal agent trained by an llm from a parallel textworld. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 26275–26285, 2024.
- [88] Ziqi Yang, Xuhai Xu, Bingsheng Yao, Ethan Rogers, Shao Zhang, Stephen Intille, Nawar Shara, Guodong Gordon Gao, and Dakuo Wang. Talk2care: An llm-based voice assistant for communication between health-care providers and older adults. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(2):1–35, 2024.
- [89] Da Yin, Faeze Brahman, Abhilasha Ravichander, Khyathi Chandu, Kai-Wei Chang, Yejin Choi, and Bill Yuchen Lin. Agent lumos: Unified and modular training for open-source language agents. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12380–12403, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.670. URL <https://aclanthology.org/2024.acl-long.670/>.
- [90] Siyu Yuan, Kaitao Song, Jiangjie Chen, Xu Tan, Yongliang Shen, Kan Ren, Dongsheng Li, and Deqing Yang. EASYTOOL: Enhancing LLM-based agents with concise tool instruction. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 951–972, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.44. URL <https://aclanthology.org/2025.naacl-long.44/>.
- [91] Xin Zeng, Xiaoyu Wang, Tengxiang Zhang, Chun Yu, Shengdong Zhao, and Yiqiang Chen. Gesturegpt: Toward zero-shot free-form hand gesture understanding with large language model agents. *Proceedings of the ACM on Human-Computer Interaction*, 8(ISS):462–499, 2024.
- [92] Simon Zhai, Hao Bai, Zipeng Lin, Jiayi Pan, Peter Tong, Yifei Zhou, Alane Suhr, Saining Xie, Yann LeCun, Yi Ma, et al. Fine-tuning large vision-language models as decision-making agents via reinforcement learning. *Advances in neural information processing systems*, 37:110935–110971, 2024.
- [93] Ceyao Zhang, Kaijie Yang, Siyi Hu, Zihao Wang, Guanghe Li, Yihang Sun, Cheng Zhang, Zhaowei Zhang, Anji Liu, Song-Chun Zhu, et al. Proagent: building proactive cooperative agents with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 17591–17599, 2024.
- [94] Danyang Zhang, Lu Chen, Situo Zhang, Hongshen Xu, Zihan Zhao, and Kai Yu. Large language models are semi-parametric reinforcement learning agents. *Advances in Neural Information Processing Systems*, 36:78227–78239, 2023.
- [95] Hongxin Zhang, Weihua Du, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, and Chuang Gan. Building cooperative embodied agents modularly with large language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=EnXJfQqyOK>.
- [96] Jintian Zhang, Xin Xu, Ningyu Zhang, Ruiibo Liu, Bryan Hooi, and Shumin Deng. Exploring collaboration mechanisms for LLM agents: A social psychology view. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14544–14607, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.782. URL <https://aclanthology.org/2024.acl-long.782/>.
- [97] Qian Zhang, Yongxu Hu, Jiaxin Yan, Hengyue Zhang, Xinyi Xie, Jie Zhu, Huchao Li, Xinxin Niu, Liqiang Li, Yajing Sun, et al. Large-language-model-based ai agent for organic semiconductor device research. *Advanced Materials*, 36(32):2405163, 2024.
- [98] Ruichen Zhang, Hongyang Du, Yinqiu Liu, Dusit Niyato, Jiawen Kang, Zehui Xiong, Abbas Jamalipour, and Dong In Kim. Generative ai agents with large language model for satellite networks via a mixture of experts transmission. *IEEE Journal on Selected Areas in Communications*, 2024.

- [99] Shaokun Zhang, Jieyu Zhang, Jiale Liu, Linxin Song, Chi Wang, Ranjay Krishna, and Qingyun Wu. Offline training of language model agents with functions as learnable weights. In *Proceedings of the 41st International Conference on Machine Learning*, ICML’24. JMLR.org, 2024.
- [100] Shiyao Zhang, Yuji Dong, Yichuan Zhang, Terry R Payne, and Jie Zhang. Large language model assisted multi-agent dialogue for ontology alignment. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems*, pages 2594–2596, 2024.
- [101] Xinyu Zhang, Huiyu Xu, Zhongjie Ba, Zhibo Wang, Yuan Hong, Jian Liu, Zhan Qin, and Kui Ren. Privacyasst: Safeguarding user privacy in tool-using large language model agents. *IEEE Transactions on Dependable and Secure Computing*, 21(6):5242–5258, 2024.
- [102] Yikai Zhang, Siyu Yuan, Caiyu Hu, Kyle Richardson, Yanghua Xiao, and Jiangjie Chen. TimeArena: Shaping efficient multitasking language agents in a time-aware simulation. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3894–3916, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.215. URL <https://aclanthology.org/2024.acl-long.215/>.
- [103] Yuge Zhang, Qiyang Jiang, XingyuHan XingyuHan, Nan Chen, Yuqing Yang, and Kan Ren. Benchmarking data science agents. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5677–5700, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.308. URL <https://aclanthology.org/2024.acl-long.308/>.
- [104] Yusen Zhang, Ruoxi Sun, Yanfei Chen, Tomas Pfister, Rui Zhang, and Serkan Arik. Chain of agents: Large language models collaborating on long-context tasks. *Advances in Neural Information Processing Systems*, 37:132208–132237, 2024.
- [105] Qinlin Zhao, Jindong Wang, Yixuan Zhang, Yiqiao Jin, Kaijie Zhu, Hao Chen, and Xing Xie. Competeai: understanding the competition dynamics of large language model-based agents. In *Proceedings of the 41st International Conference on Machine Learning*, ICML’24. JMLR.org, 2024.
- [106] Sipeng Zheng, jiazheng liu, Yicheng Feng, and Zongqing Lu. Steve-eye: Equipping LLM-based embodied agents with visual perception in open worlds. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=NltzxpG0nz>.
- [107] Hang Zhou, Yehui Tang, Haochen Qin, Yujie Yang, Renren Jin, Deyi Xiong, Kai Han, and Yunhe Wang. Star-agents: Automatic data optimization with llm agents for instruction tuning. *Advances in Neural Information Processing Systems*, 37:4575–4597, 2024.
- [108] Feiyu Zhu and Reid Simmons. Bootstrapping cognitive agents with a large language model. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 655–663, 2024.